

Day – 8 Network Troubleshooting, Management, Monitoring and Optimization.

Task – 1 Use windows command to config ip address.

- ? Press the Windows Key, type cmd, and press Enter to open the Command Prompt.
- ? Type the following command and press Enter:
- DOS
- ipconfig /all
- ? Locate your active network adapter (such as Wireless LAN adapter Wi-Fi or Ethernet adapter) and look for these four specific lines:

Task – 2 Network connectivity issue .

- 1. When Disconnected (Wi-Fi Off or Cable Unplugged)
- What happens when you run ping www.google.com: Because your computer has no active link to a local network or the internet, it cannot send packets anywhere.
- The Result: You will see error messages such as "General failure", "Destination Host Unreachable", or "Request timed out."
- What it means: The operating system's network stack recognizes that the physical layer is down or that no default gateway is available to forward traffic outside your machine.
- 2. When Reconnected (Wi-Fi Back On)
- What happens when you run ping www.google.com again: Your laptop re-establishes a link with your router, re-acquires its IP configuration, and opens the path to the internet.
- The Result: The command springs back to life, showing consecutive lines like Reply from 142.250...: bytes=32 time=15ms TTL=115.
- What it means: End-to-end network connectivity is fully restored. Your packets are successfully traveling across your local router, through your ISP's network, and back from Google's servers with measurable response times.

Task – 3 Task manager on the system.

- However, if you open Resource Monitor on your Windows computer (press Ctrl + Shift + Esc to open Task Manager, click on the Performance tab, and select Open Resource Monitor at the bottom, then go to the Network tab), you can easily observe this yourself.
- Here is what you will typically record during a 5-minute YouTube streaming session:
 - 1. Process with the Highest Bandwidth Usage
 - Process Name: chrome.exe (or msedge.exe, depending on which browser you use).
 - Why: The browser process handling the active YouTube tab will consume the vast majority of your network throughput as it continuously downloads video and audio chunks (buffers data ahead of playback).
 - 2. Typical Peak Network Usage Values
 - For 1080p Streaming: Peak bandwidth usage typically spikes between 3 Mbps to 8 Mbps (kilobytes to megabytes per second depending on whether it's loading initial chunks or playing back smoothly).
 - For 4K Streaming: Peak bandwidth can jump anywhere from 15 Mbps to 25 Mbps+ during heavy initial buffering or high-resolution playback spikes.

Task – 4 Based on scenario.

- 1. TLS/SSL Handshake Parameters (for HTTPS)
 - What you see: Before any encrypted data flows, you can inspect the initial Client Hello and Server Hello messages. This reveals supported cipher suites, TLS version details (such as TLSv1.3), extensions, and the cryptographic parameters negotiated between your browser and the web server.
- 2. Destination IP Addresses and Server Hostnames
 - What you see: You can identify the exact public IP addresses of the cloud servers or content delivery network (CDN) nodes hosting the platform. If inspecting plain HTTP or unencrypted headers/SNI (Server Name Indication) fields in TLS, you can clearly trace domain names and routing endpoints.
- 3. TCP Connection Management and Packet Flow
 - What you see: You can observe the underlying transport mechanics, including the initial three-way handshake (SYN, SYN-ACK, ACK), sequence numbers, acknowledgment numbers, window sizes, and connection teardown flags (FIN or RST) that manage session reliability.

Task – 5 Based on scenario.

- Home Network Gaming Optimization Checklist:
- Switch to a Wired Ethernet Connection: Bypass wireless interference and packet loss by plugging your gaming device directly into the router with an Ethernet cable.
- Enable Quality of Service (QoS): Configure your router's QoS settings to prioritize gaming packets over background downloads, streaming, or general web browsing.
- Switch to the 5 GHz (or 6 GHz) Wi-Fi Band: If gaming on wireless is unavoidable, separate your SSIDs and ensure your device is connected to the less congested, higher-speed 5 GHz band rather than the crowded 2.4 GHz channel.
- Power Cycle Network Hardware: Restart both your modem and router to clear out temporary memory caches, flush stale connections, and reduce local latency.
- Update Router Firmware: Log into your router's administrative dashboard to ensure the device is running the latest manufacturer firmware, which patches performance bugs and enhances stability.
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